BIJO SEBASTIAN

Email: bijo.sebastian@iitm.ac.in
LinkedIn Profile | Github

I am an Assistant Professor in the Department of Engineering Design and current head of the Autonomous Systems Laboratory at the Indian Institute of Technology, Madras. Before joining IIT Madras, I was a Product Owner and Software Engineer with the Autonomy Team of Torc Robotics, based out of Virginia USA, where I worked on developing motion planning algorithms for Daimler Freightliner trucks.

I received my B.Tech degree in Mechanical Eng. from College of Engineering, Trivandrum in 2013, MS in Mechatronics from Central Mechanical Engineering Research Institute, West Bengal in 2015 and Ph.D. in Mechanical Engineering with focus on Robotics form the Virginia Polytechnic Institute and State University in 2019.

My research interests include developing intelligent grasping and manipulation algorithms, developing autonomous navigation solutions for mobile robots and multi-agent systems. I am a member of the Institute of Electrical and Electronics Engineers (IEEE), Institution of Engineers (IEI)-India, The Robotics Society (TRI) - India and the American Society of Mechanical Engineers (ASME). For more details on ongoing and prior research please refer to my Lab website.

Current Role

Assistant Professor November-2021-Present
Department of Engineering Design,
Indian Institute of Technology Madras
Tamil Nadu, India

Industrial Experience

Product Owner January 2021-November-2021
Behaviors, Planning and Controls Team
Torc Robotics, Virginia, USA

Software Engineer November 2019-January 2021
Motion Planning Team
Torc Robotics, Virginia, USA

Education

Ph.D. Mechanical Engineering January 2016-October 2019
Virginia Tech, USA
Dissertation: Traversability estimation techniques for improved navigation of tracked mobile robots

M.Tech, Mechatronics August 2013-August 2015
Academy of Scientific and Innovative Research, India
Thesis: Design, analysis, and development of climbing robot
Second rank in 2015 Mechatronics batch of four students from CMERI

B.Tech, Mechanical Engineering August 2009 - May 2013
University of Kerala, India

Best outgoing student from College of Engineering Trivandrum
Fourth rank in Kerala University, 2013 ME Batch

Research Experience

Software Engineer, Autonomy stack November 2019 – November2021

Torc Robotics, Virginia, USA

- As a part of the Autonomy Team, I was responsible for software development along with engineering activities to support delivery of the Level 4 Autonomy Stack as part of the overall system solution. From a software development standpoint this involves developing algorithms to handle high level decision making, path planning and motion control while considering the perception data about the environment as obtained from the sensors in real-time. From an engineering standpoint this involves identifying benchmark tests, deploying, and testing the algorithm performance in real world conditions and adding new capabilities to meet the autonomous navigation requirements.

Graduate Research Assistant January 2016-October 2019

Robotics and Mechatronics Lab, Virginia Tech,
USA

- Traversability estimation techniques for improved navigation of tracked mobile robots: Tracked robotic systems are widely used for search and rescue applications in challenging terrain conditions. For this use case, I have explored the use of Physics Engines to aid in path planning, developed Active Disturbance Rejection Control strategy to address vehicle slip and developed Support Vector Machine based architecture for real-time terrain identification.
- Semi-Autonomous Victim Extraction Robot: Developed conceptual design for a robotic system capable of extracting an injured person from a disaster or war scenario and then evacuate them

to a safe location for further medical attention. Developed simulations demonstrating casualty extraction capability using the proposed design. A US provisional patent has been filed on this design.

- Development of a haptic exoskeleton glove: Design, integration of two different versions of exoskeleton gloves, which are lightweight, portable and acts as self-contained mechatronic systems. Optimized underlying mechanism to imitate natural finger motion, designed and developed miniature series elastic actuation systems to enable precise force control on each finger. A US patent has been filed on the developed design.

Project Fellow June 2014-November 2015

Surface Robotics Lab, CSIR-CMERI, India

- Design, analysis, and development of climbing robot: The proposed system was developed for horticultural and pipeline inspection applications. A novel design was developed and validated through ADAMS–MATLAB Co–Simulation. A complete prototype was integrated with actuators, sensors, and necessary control software to make the prototype fully operational. Experimental validation of the proposed design and climbing strategies were also performed.
- Validation of Navigational algorithms on PIONEER P3-AT platform: Developed and tested various state of the art algorithms for path planning and localization along with control strategies for obstacle avoidance and goal seeking to create a complete navigation module. Implemented and tested these algorithms on a simulated PIONEER P3-AT as well as a real ROVER 5 platform with integrated sensing and onboard processing.
- Automated mechanism for plate washing: A novel plate washing mechanism specifically designed for removing oily food waste from plates. The proposed system was developed into a prototype and a design registration has been filed on the same under CSIR.
- Design of a spherical robot with simplified drive mechanism: A modified pendulum drive system was designed for a spherical robot. In addition to being completely self-contained, the proposed design is much simpler and more versatile than the existing spherical robot designs.

Selected publications

- **Sebastian, B.,** Ben-Tzvi, P., "Physics Based Path Planning for Autonomous Tracked Vehicle in Challenging Terrain" *Journal of Intelligent and Robotic Systems*, Springer, Vol. 95, Issue 2, pp. 511-526, August 2019. DOI: 10.1007/s10846-018-0851-3
- **Sebastian, B.,** Ben-Tzvi, P., "Active disturbance rejection control for handling slip in tracked vehicle locomotion", *Journal of Mechanisms and Robotics*, Transactions of the ASME, Vol.11, Issue 2, pp. 021003:1-12, April 2019 DOI: 10.1115/1.4042347
- **Sebastian, B.,** Ben-Tzvi, P., "Support Vector Machine Based Real-time Terrain Estimation for Tracked Robots", *Mechatronics*, Elsevier, Vol. 62, pp. 102260, October 2019. DOI: 10.1016/j.mechatronics.2019.102260
- Chauhan, R., **Sebastian, B.,** and Ben-Tzvi, P., "Grasp Prediction Toward Naturalistic Exoskeleton Glove Control," *IEEE Transactions on Human-Machine Systems*, vol. 50, no. 1, pp. 22-31, February 2020, DOI: 10.1109/THMS.2019.2938139

- Williams, A., **Sebastian, B.,** Ben-Tzvi, P., "A Robotic Head Stabilization Mechanism for Medical Transport", *Robotics*, Multidisciplinary Digital Publishing Institute, Vol. 8, Issue 1, pp. 23, March 2019. DOI: 10.3390/robotics8010023
- Williams, A., **Sebastian, B.,** & Ben-Tzvi, P., "Review and Analysis of Search, Extraction, Evacuation, and Medical Field Treatment Robots", *Journal of Intelligent and Robotic Systems,* Springer, Vol. 96, pp. 401–418 February 2019. DOI: 10.1007/s10846-019-00991-6
- Refour, E. M., **Sebastian, B.,** Chauhan, R. J., and Ben-Tzvi, P. "A General Purpose Robotic Hand Exoskeleton With Series Elastic Actuation." *Journal Mechanisms and Robotics*. December 2019; 11(6): 060902.DOI: 10.1115/1.4044543

For the complete list of publication please refer to my Google Scholar profile.

Patents

- Ben-Tzvi, P., **Sebastian, B.,** Refour, E., Xu, W., Pradhan, S., Guo, Y., "Robotic exoskeleton glove system," U.S. Patent No.11986033, May 21, 2024
- **Sebastian, B.,** "Bidirectional Rotary Series Elastic Actuator (R-SEA) Design Usin Coil Compression Springs," Indian Patent No.544679, August 25, 2023
- Patel, N., **Sebastian, B.,** "Compound Offset Epicyclic Drive (COED)," Indian Patent No.579416, July 28, 2023

Honors and Awards

- **Best Paper Award** for our work titled "Intelligent Grasp Planning using Open-Vocabulary Image Segmentation and Large Language Models" presented at the IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control (STPEC 2025) at NIT, Goa, 10 – 13 December, 2025.
- Award for **Excellence in Teaching** for the year 2025, Indian Institute of Technology Madras, Chennai, India
- **Best Presentation Award** at the IEEE ICCAR 2024 Conference (27-29 April, 2024) for our work titled "Shared Control for Telepresence Robot in a Dynamic Virtual Reality Environment."
- Ideation contest 2014 - **Participation prize** for "Detailed design of novel spherical robot", TePP Outreach cum Cluster Innovation Center, CSIR-CMERI, Durgapur, India
- Student Mechanism Design Contest - **First Prize** for "Design and integration of novel mechanism for plate washing", iNaCoMM 2013, IIT Roorkee, India
- Col. P. Vaidyanathan Memorial Endowment and K. Thomas Kora Memorial Award - **Best Outgoing Mechanical Engineering Student** College of Engineering, Thiruvananthapuram, India; March 2014

Leadership and Service

- **Coordinator for Inclusive Education, Dept. of Engineering Design, IIT Madras:** IIT Madras' Inclusive Education initiative seeks to nurture students with disabilities (SwD) such that they become professionally competent and personally independent, by enabling reasonable participation in curricular, co-curricular and extra-curricular activities in the IITM campus. Since 2023

- **Seminar:** Invited talk on the topic “AI & Data: Engines of Future Mobility” at Bajaj Auto Technology Limited, India. 11th November 2025
- **Seminar:** Invited talk on the topic “Intelligent grasping and manipulation for modern industry”, Accenture Robotics Community Call, 28th August, 2025
- **Seminar:** Invited talk on the topic “Advances in Mobile Robotics and Multi-Agent Systems: Applications in Modern Industry”, Caterpillar AI Community of Practice, 4th September, 2025
- **Seminar:** Invited talk on the topic “Design, modelling and control or mobile robots and manipulators” at St. Josephs College, chennai, India. 22nd February 2025
- **Seminar:** Invited talk on the topic “Design, Integration, and Validation of robotic systems to address challenges in modern industry” at Accenture Innovation Labs, Bangalore India. 20th February 2024
- **Seminar:** Invited talk on the topic “Design, modelling, and Control of Robotic Systems: Exoskeletons & Mobile Robots” at College of Engineering Trivandrum Kerala, India. 4th March 2024
- **Seminar:** Invited talk on the topic “Design, modelling and control of robotics systems : Mobile robots and Manipulators” to Sri Sivasubramaniya Nadar (SSN) College of Engineering, Kalavakkam. 18th March 2024
- **Seminar:** Invited talk on the topic “Design, motion planning and control of robotic systems: Mobile robots and Manipulators” to Rajalakshmi Engineering College, Chennai. 29th May 2024
- **Seminar:** Invited talk on the topic “Application of Artificial Intelligence in Industrial Robotics" on 3rd July 2024 at Continuing Education Program (CEP) Heavy Vehicle Factory, Avadi, Chennai.
- **Seminar:** Invited talk on "Smart Automation: AI and Robotics for Industrial Needs" as part of workshop conducted at National Institute of Technology Tiruchirappalli, Tamil Nadu on 7th November 2024
- **Webinar:** On the topic “2D LiDAR based SLAM using ROS” to the engineers of Samsung R&D Institute India – Delhi on 14th November 2022
- **Seminar:** On the topic “Introduction to Robotics” to Master students within the Department of Physics at Newman College, Kerala India. 2nd August 2017
- **Robotics Workshop:** Including lectures and hands-on sessions as part of a two-day robotics workshop for undergraduate students. February 2015
- **STEP 4 U Instructor:** Volunteered under Kerala state government sponsored program and took support lectures on computer programming for undergraduate students. 2011-2013